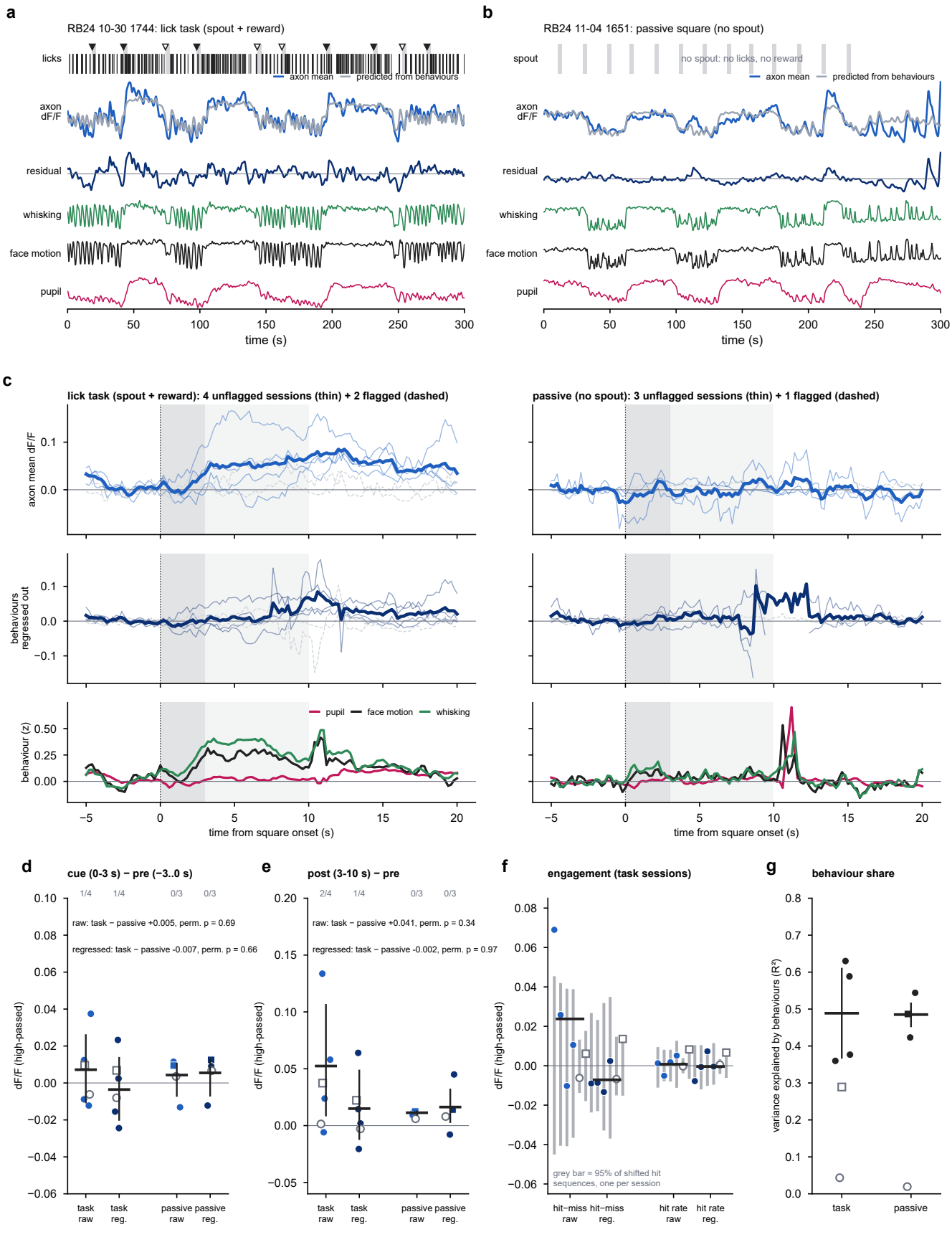



a, b: ▼ = square onset (task: filled hit, open miss), grey = 3 s after onset; axon mean and its prediction from pupil + face + whisking + ball, residual = difference. Signal: mean best dF/F of the signal-like ROIs, high-passed > 30 s. c-g: behaviours (pupil, face, whisking, ball where plausible) at lags -1..+3 s enter one ridge model together with the epoch regressors pre (-3..0 s), cue (0-3 s) and post (3-10 s) after square onset (reference: rest of the cycle); raw = epochs only, regress. = with behaviours. Null: epoch pattern circularly shifted by >= 60 s (500 draws); n/N = unflagged sessions with z > 2.33. c: residual = axon mean minus the behaviour part of the model. Task: 1752, 1744, 1423, 1520 (RB24) + flagged 1317, RB23 1704; passive: 1456, 1651 (RB24), RB23 1423 + flagged 1529. f: hit-trial cycle minus miss-trial cycle and hit rate of the surrounding 10 trials (per SD), same model, null = hit sequence shifted by >= 5 trials. Black = mean of per-animal means, 95% hierarchical bootstrap CI (task: one animal); task and passive differ in animal and training day. In 1456, 1752 and RB23 1423 / 1704 a pupil / face transient follows the square at ~11 s (end of grey).